Medical Microbiology

Lecture Eleven

Mycobacterium Tuberculosis

Third stage- College of Dentistry

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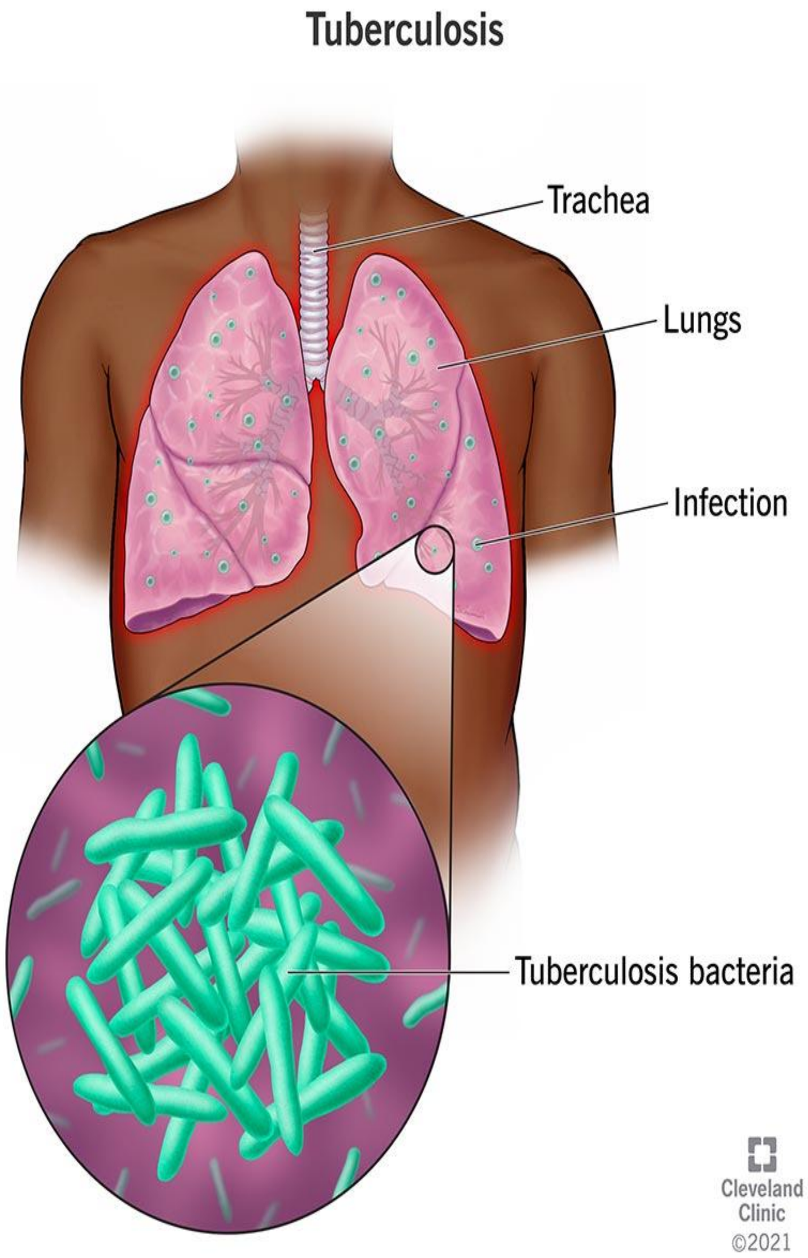
Introduction:

- ***Mycobacterium tuberculosis:*** First discovered in 1882 by Robert Koch, possesses a distinctive waxy cell wall rich in mycolic acids.
- This unique structure renders the bacterium resistant to conventional Gram staining, causing it to appear weakly Gram-positive .
- Consequently, acid-fast staining techniques, such as the **Ziehl Neelsen stain**, or **fluorescent staining** methods, are employed for reliable microscopic identification of *M. tuberculosis.*

- Mycobacterium tuberculosis is the causative agent of tuberculosis (TB), a chronic infectious disease primarily affecting the lungs but capable of spreading to other organs.
- Despite being an ancient disease, TB remains a global health concern, particularly in developing countries.

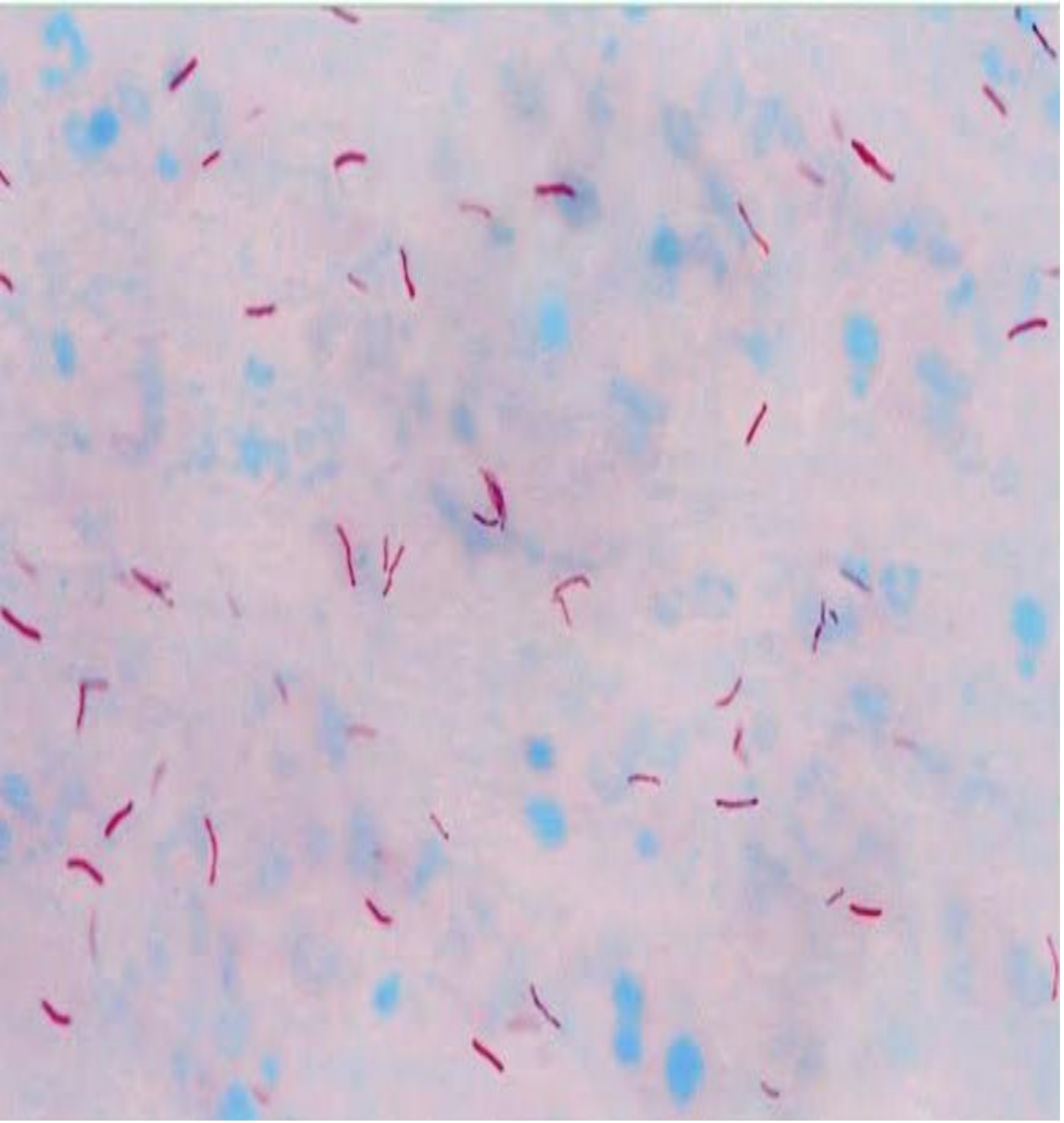


Morphology and Characteristics

- Mycobacterium tuberculosis is a slender, rod-shaped, acid-fast bacillus characterized by a waxy cell wall rich in mycolic acids.
- The organism is small, resistant to weak disinfectants, and capable of remaining viable in a dry environment for weeks.
- Its lipid-rich cell wall, composed of mycolic acids and cord factor glycolipids, is responsible for desiccation resistance and represents an important virulence determinant

Morphology and Characteristics

- It is an obligate aerobe, meaning it thrives in oxygen-rich environments such as the lungs.
- The bacterium exhibits slow growth, requiring 14–28 days to grow on selective media.
- The bacterium is non-motile, non-spore-forming, non-capsulated, and does not produce toxins.



Virulence Factors

M. tuberculosis employs various mechanisms to evade the immune system and establish infection includes:

1. Mycolic Acid and Cell Wall Components: The thick lipid-rich cell wall provides resistance to desiccation, antibiotics, and host immune responses.

2. Cord Factor (Trehalose Dimycolate): Inhibits macrophage function and contributes to granuloma formation.

3. Lipoarabinomannan (LAM): Suppresses immune responses and promotes survival within macrophages.

4. ESX-1 Secretion System: This system helps the bacteria to transport proteins across the thick cell wall and thus facilitates escape from phagosomes and enhances virulence.

5. Catalase-Peroxidase (KatG): Helps the bacterium resist oxidative stress.

Important Medical Note: This enzyme is responsible for activating the drug Isoniazid, which is one of the most essential medications for treating tuberculosis. If the bacteria lose this enzyme through mutations, they become resistant to this drug.

Q/Which mechanism does *M. tuberculosis* utilize to transport proteins across its thick cell wall, thereby facilitating its escape from phagosomes to enhance its virulence?

A) Cord Factor (Trehalose Dimycolate)

B) ESX-1 Secretion System

C) Catalase-Peroxidase (KatG)

D) Mycolic Acid barrier

- **Correct Answer:**

B) ESX-1 Secretion System

- **Why B is correct:** The provided documents state that the **ESX-1 Secretion System** helps the bacteria transport proteins across the thick cell wall, which facilitates the escape from phagosomes and enhances overall virulence.
- **Why others are incorrect:**
- **Cord Factor** is linked to inhibiting macrophage function and contributing to granuloma formation.
- **KatG** is an enzyme that protects against oxidative stress and is not a transport system.
- **Mycolic Acid** is a structural component of the wall itself rather than a secretion system.

Pathogenesis and Clinical Manifestations:

1. **Primary Infection:** Inhaled droplet nuclei containing M. tuberculosis reach the alveoli, where they are engulfed by alveolar macrophages. If not eliminated, the bacteria multiply within macrophages.

2. **Granuloma Formation:** The immune response leads to the formation of tubercles (granulomas) that contain the bacteria.

3. **Latent TB Infection (LTBI):** The bacteria remain dormant within granulomas, with no active symptoms but potential for reactivation.

4. **Active TB Disease:** When immunity weakens (e.g., HIV, malnutrition, diabetes) the bacteria spread, causing pulmonary symptoms like chronic cough, hemoptysis, fever, night sweats, and weight loss.

5. **Extra pulmonary TB:** Can affect lymph nodes, bones, kidneys, and the CNS (e.g., tuberculous meningitis).

Question: A patient with a history of latent TB infection (LTBI) is diagnosed with poorly controlled diabetes. Recently, they have developed a chronic cough, night sweats, and hemoptysis. Which of the following best explains the progression of their condition?

A) The bacteria were initially eliminated by alveolar macrophages during primary infection.

B) A weakened immune system allowed dormant bacteria within granulomas to multiply and spread.

C) The ESX-1 secretion system prevented the bacteria from entering the alveoli initially.

D) The presence of Lipoarabinomannan (LAM) triggered an immediate symptomatic pulmonary response.

- **Solution and Explanation:- Correct Answer: (B)**
- **Rationale:** In Latent TB Infection (LTBI), bacteria remain dormant within granulomas with no active symptoms. When immunity weakens—specifically due to conditions like **diabetes**, HIV, or malnutrition—the bacteria can spread, leading to "Active TB Disease" characterized by symptoms like chronic cough, hemoptysis, and night sweats.
- **Why others are incorrect:**
- **A:** If bacteria were eliminated during primary infection, the patient would not have developed LTBI or a subsequent active disease.
- **C:** The **ESX-1 system** facilitates the escape from phagosomes after infection; it does not prevent entry into the alveoli.
- **D:** While **LAM** suppresses immune responses to promote survival, the symptoms described are characteristic of the "Active TB Disease" stage following a period of dormancy, not an immediate response.

Diagnosis:

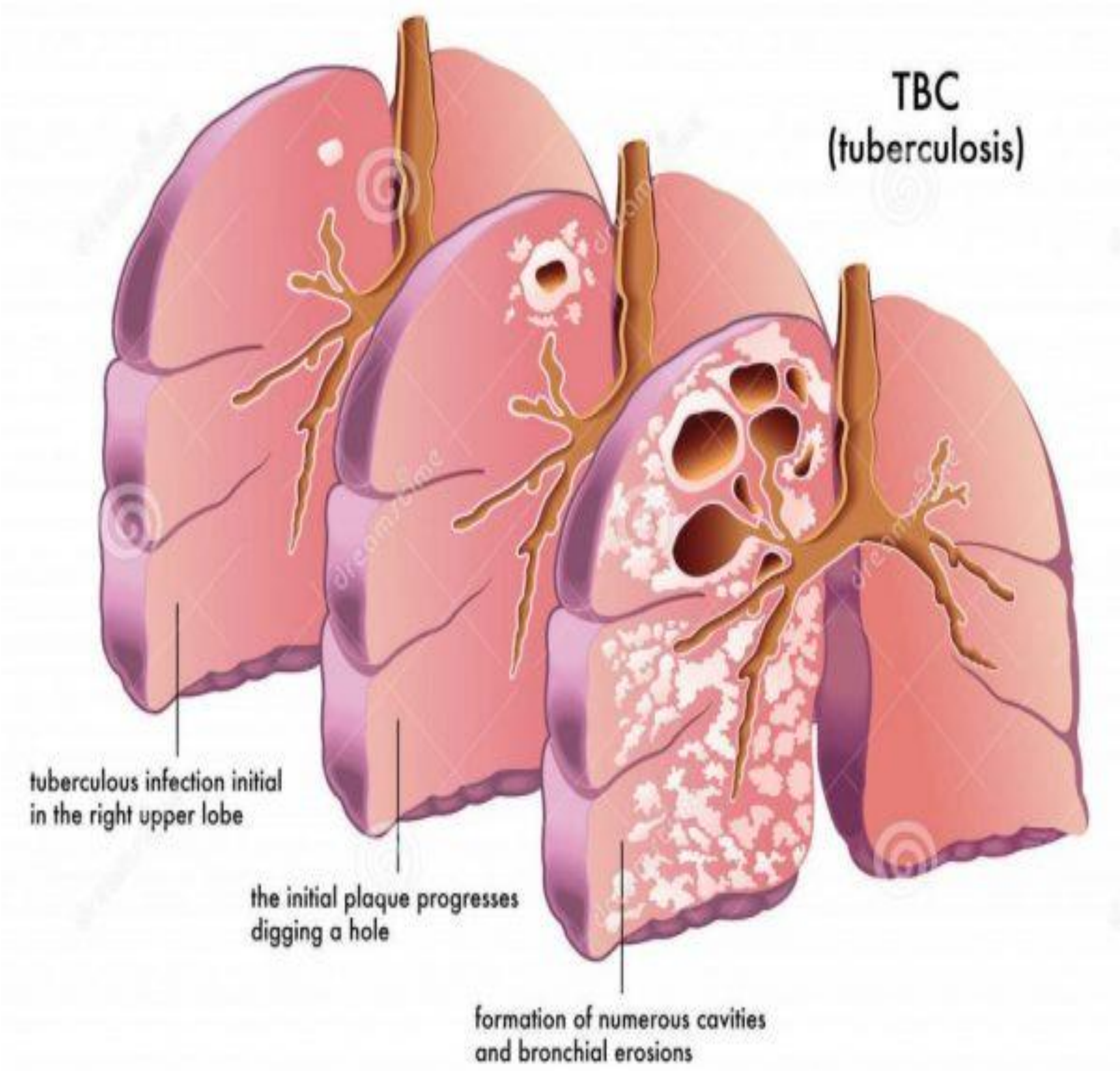
1. Microscopy: Acid-fast staining (Ziehl-Neelsen stain) detects mycobacteria in sputum samples.

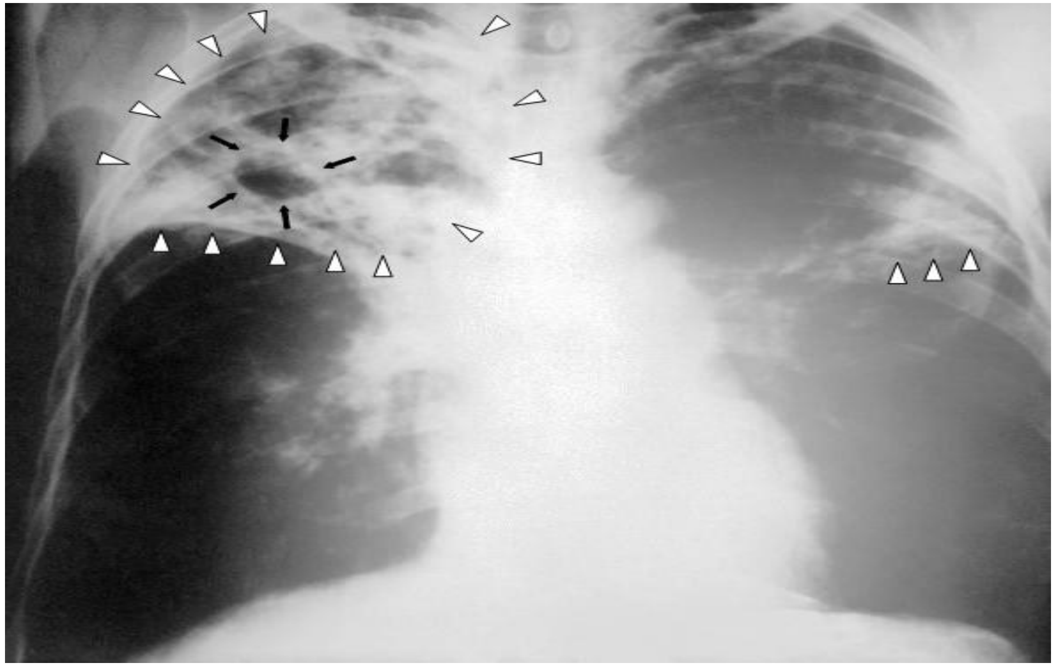
2. Culture: Lowenstein–Jensen agar supports bacterial growth, but colony development requires several weeks.

3. Molecular Tests: Molecular diagnostic methods based on PCR, such as the GeneXpert assay, enable rapid detection of *M. tuberculosis*.

4. Tuberculin Skin Test (TST) to Identify latent TB infection.

5. Radiology: Chest X-rays and CT scans show pulmonary lesions.


TBC
(tuberculosis)
tuberculous infection initial
in the right upper lobe
the initial plaque progresses
digging a hole
formation of numerous cavities
and bronchial erosions



Prevention and Control:

- **BCG Vaccine**: Provides partial protection, especially in children.
- **Infection Control**: Proper ventilation, respiratory hygiene, and isolation of contagious patients in healthcare settings.
- **Public Health Measures**: Screening high-risk populations and ensuring treatment adherence to prevent spread.

Treatment of Tuberculosis (TB):

The treatment of tuberculosis requires **prolonged combination antibiotic therapy** to prevent drug resistance and ensure complete eradication of *Mycobacterium tuberculosis*.

- **First-line anti-tubercular drugs include:**
- **Isoniazid (INH)**
- **Rifampicin**
- **Ethambutol**
- **Pyrazinamide**
- **Standard treatment regimen (drug-susceptible TB):**
- **Intensive phase:**
  Isoniazid + Rifampicin + Pyrazinamide + Ethambutol

  for **2 months**
- **Continuation phase:**
  Isoniazid + Rifampicin for **4 months**

Conclusion:

- Tuberculosis remains a significant global health challenge.
- Understanding its pathogenesis, diagnosis, and treatment is crucial for healthcare professionals, including dentists, as they may encounter oral manifestations of TB such as ulcerations, lymphadenopathy, and osteomyelitis of the jaw.
- Fast identification and referral can aid in early diagnosis and control of the infection.

Any Question?

Thank you